Sprint 2: Analysis

TEMA: A Qualitative Analysis Application

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Section 1: Refined Requirements

Functional Requirements

Note: Our users are qualitative researchers looking for a visual workflow

FR1 **Display Start Page:** The user should see a start page upon launching the application.
- FR1.1 **Create Canvas:** The user can create a new canvas from the start page.
- FR1.2 **View Created Canvases:** The user can open a list of previously made canvases.
  - FR1.2.1 **Open Created Canvas:** The user can select a created canvas from this list to open it within the application.
  - FR1.2.2 **Delete Created Canvas:** The user can delete a created canvases from this list.
- FR1.3 **Import Created Canvases:** The user can import a created canvas using its JSON save data.

FR2 **Send to Qualitative Analysis App:** The user must be able to send all qualitative codes to an opened canvas.
- FR2.1 **Send Google Doc Codes:** The user can send all qualitative codes within a Google Doc's comments section to an opened canvas within Google Docs.
- FR2.2 **Send Google Sheet Codes (Button):** The user can send all qualitative codes within a Google Sheet to an opened canvas within Google Sheets.
- FR2.3 **Send Google Sheets Codes (Copy Paste):** The user can copy a selection of cells in a Google Sheet and paste them into an opened canvas.

FR3 **Convert to Sticky Notes:** The system must convert each qualitative code sent via Google Doc or Google Sheets into its own sticky note.
- FR3.1 **Make User Select Canvas:** If a canvas is not opened in the application, the system shall ask the user if they want to make a new canvas for the sticky notes or to send them to an existing canvas.
- FR3.2 **Drop Sticky Notes into Canvas:** The system shall take the sticky notes gotten from Google Docs or Google Sheets and display them in the canvas in a grid pattern.

FR3.2.1 **Check for Existing Sticky Notes:** The system shall check for sticky notes already within the canvas to make sure the newly inserted sticky notes do not overlap with established sticky notes.

FR4 **Create Individual Sticky Notes:** The user can create an individual sticky note.

FR5 **Move Sticky Notes:** The user can drag and drop the sticky notes around the canvas.

FR6 **Edit Sticky Notes:** The user can edit a sticky note's text contents, color, size, etc.

FR6.1 **Edit Sticky Note Text:** The user can edit sticky note text.

FR6.2 **Edit Sticky Note Color:** The user can edit the color of a sticky note.

FR6.3 **Edit Sticky Note Size:** The user can edit the size of the sticky note.

FR7 **Group Sticky Notes:** The user can assign sticky notes to distinct groups.

FR7.1 **Designate Non-Overlapping Groups:** The user can designate groups as non-overlapping. These groups cannot share sticky notes or groups with any other group.

FR7.2 **Designate Overlapping Groups:** The user can assign groups as overlapping. These groups can share sticky notes and groups with other overlapping groups.

FR7.2.1 **Designate Overlapping by Default:** The system will make any newly created group overlapping by default.

FR7.3 **Color-Code Groups:** The user can assign groups different colors. Sticky notes will change their background color based on their group's colors

FR7.3.1 **Color Overlapping Groups:** The system will display the background of a sticky note assigned to multiple groups as a split between these groups' colors.

FR8 **Change Sticky Note Group:** The user must be able to reassign a sticky note to different groups.

FR9 **Add Hierarchy:** The user canestablishhierarchies to the created groups, allowing for layers of subgroups within larger groups

FR10 **Change Hierarchy:** The user can change the hierarchy level of a group to a higher or lower level.

FR11 **Display Statistics:** The user can view statistics about the canvas and groups.

FR11.1 **Display Number of Codes:** The system can show the number of codes.

FR11.1.1 **Display Codes per Group:** The system can show the number of codes per group.

FR11.2 **Display Number of Groups:** The system can show the number of groups.

FR11.2.1 **Display Number of Subgroups per Group:** The system can show the number of subgroups within a group.

FR11.3 **Display Dendrogram:** The system can display each hierarchy within the canvas as a dendrogram.

FR12 **Export Results:** The user can export the canvas.

FR12.1 **Export as PNG:** The system will take a picture of the entire canvas to save as a PNG.

FR12.2 **Export as JSON:** The system will export the canvas in a JSON format.

FR12.3 **Export as CSV:** The system will export the canvas as a CSV.

FR13 **Refresh Data:** The user can refresh data in the canvas to see newly added codes or groups.

FR14 **Save Canvas:** The user can save their canvases.

FR15 **Assign Default Keyboard Shortcuts:** The system will have a list of predefined keyboard shortcuts that the user can use.

FR15.1 **Change Keyboard Shortcuts:** The user can change keyboard shortcuts.

Non-Functional Requirements

NFR1 **Google Docs Code Extraction [Usability]:** The user must be able to extract codes from comments in Google Docs with one click after activating our extension.

NFR2 **Google Sheets Code Extraction [Usability]:** The user must be able to extract codes from cells within a Google Sheets with two clicks after activating our extension.

NFR3 **Displaying Codes [Responsiveness]:** The UI must remain responsive up to having 500 codes on the user's screen.

NFR4 **Deployment Mode [Portability]:** The app must run locally on a single machine without any calls to external servers either directly or through APIs (e.g., checking Google data) if the Google extension is not active.

NFR5 **In-App Sticky Notes Editing [Usability]:** Changes to sticky notes must be possible entirely within the app without opening or returning to the original Google Doc or Google Sheet.

NFR6 **Confidential Data Protection [Security]:** The system must protect confidential user data including user identifiers (name, email), credentials, internal document content, and IP address. Sticky notes must be stored entirely within the local machine running the app without opening or returning to the original Google Doc or Google Sheet.

NFR7 **Sticky Notes Editing Options [Usability]:** The user can edit sticky note properties with an average of two clicks or less.

NFR8 **Group Editing Options [Usability]:** The user must be able to assign colors for groups in at most three clicks.

NFR9 **App Theming [Flexibility]:** The user can customize the app through pre-defined themes (e.g., light mode, dark mode).

NFR10 **Keyboard Shortcut Length [Efficiency]:** Predefined shortcuts for menu options and actions (e.g., assigning a code to a group) will use no more than three keys at the same time.

NFR11 **Note Auto-Sizing [Visualization]:** Sticky notes must dynamically resize based on code text length and the number of codes within a theme.

NFR12 **Group Importance Balancing [Visualization]:** The system must provide a visual mechanism to balance the emphasis of groups based on the actual number of sticky notes they contain.

Constraints

C1 **Document Access [Security]:** The system must pull codes only from documents the user explicitly authenticated or grant access to.

C2 **Workspace Compatibility [Interoperability]:** The system must integrate with Google Docs and Google Sheets.

C3 **Cross Compatibility [Implementation]:** The system must be available for Linux and Windows.

Section 2: User Stories

User Epic 1: Qualitative Researcher Coding

As a qualitative researcher, I want to conduct thematic analysis on the codes in my Google Doc so that I can draw out the overarching themes of my data.

User Story 1. As a qualitative researcher, I can click the “export” button in the TEMA extension so that my comments (codes) appear as sticky notes in an opened TEMA canvas (FR2.1; FR3)

User Story 2. As a qualitative researcher, I can create new canvases and open existing canvases in TEMA so that I can view and edit my work (FR1.1; FR1.2)

User Story 3. As a qualitative researcher, I need to be able to save my canvases so that I can resume work at a later point with no progress lost (FR14)

User Story 4. As a qualitative researcher, I can drag sticky notes around the canvas so that I can visually organize intermediate codes and themes (FR5)

User Story 5. As a qualitative researcher, I can edit and create sticky notes so that I can make changes I need (e.g., typos, additional context, etc.) (FR4; FR6)

User Story 6. As a qualitative researcher, I can (re)group sticky notes and intermediate codes into other intermediate codes so that I can build up my analysis (FR7; FR8; FR9; FR10)

User Story 7. As a qualitative researcher, I want to save my canvas as a PNG so that I can share my work in papers, on websites and in presentations (FR12.1)

User Epic 2: Peer Reviewer

As a peer reviewer, I want to see a previously made thematic analysis in TEMA so that I can examine the work done and check for issues.

User Story 1. As a qualitative researcher, I can export my canvas in the form of a JSON so that I can distribute my work with others who use the TEMA application (FR12.2)

User Story 2. As a peer reviewer, I need to take a given TEMA JSON and load it as a new canvas in the TEMA application so that I can examine the work my fellow researcher has done

(FR1.3)

User Story 3. As a peer reviewer, I want to see all the important statistics of the canvas (e.g., number of sticky codes, themes) so that I can understand the intermediate codes that led to each theme
(FR11.1; FR11.2)

User Story 4. As a peer reviewer, I want to see the structure of the analysis as a dendrogram so that I can systematically see the sticky notes and codes in each theme
(FR11.3)

User Story 5. As a peer reviewer, I want to export sticky notes, codes and themes to a CSV so that I can perform my own statistical analysis on them or replicate the work
(FR12.3)

Section 3: Product Backlog

Backlog Items

MVP Definition

The MVP of the main TEMA system is defined as the smallest functional version of the standalone application that enables qualitative researchers to organize codes into thematic groups using a visual canvas interface. Our MVP of Backlog items prioritizes this system, which primarily include a start page, sticky note manipulation & editing, and canvas creation and documentation.

The MVP for the TEMA Google extension must allow users to extract highlighted comments from a Google Doc and convert them into sticky notes inside the main TEMA application.

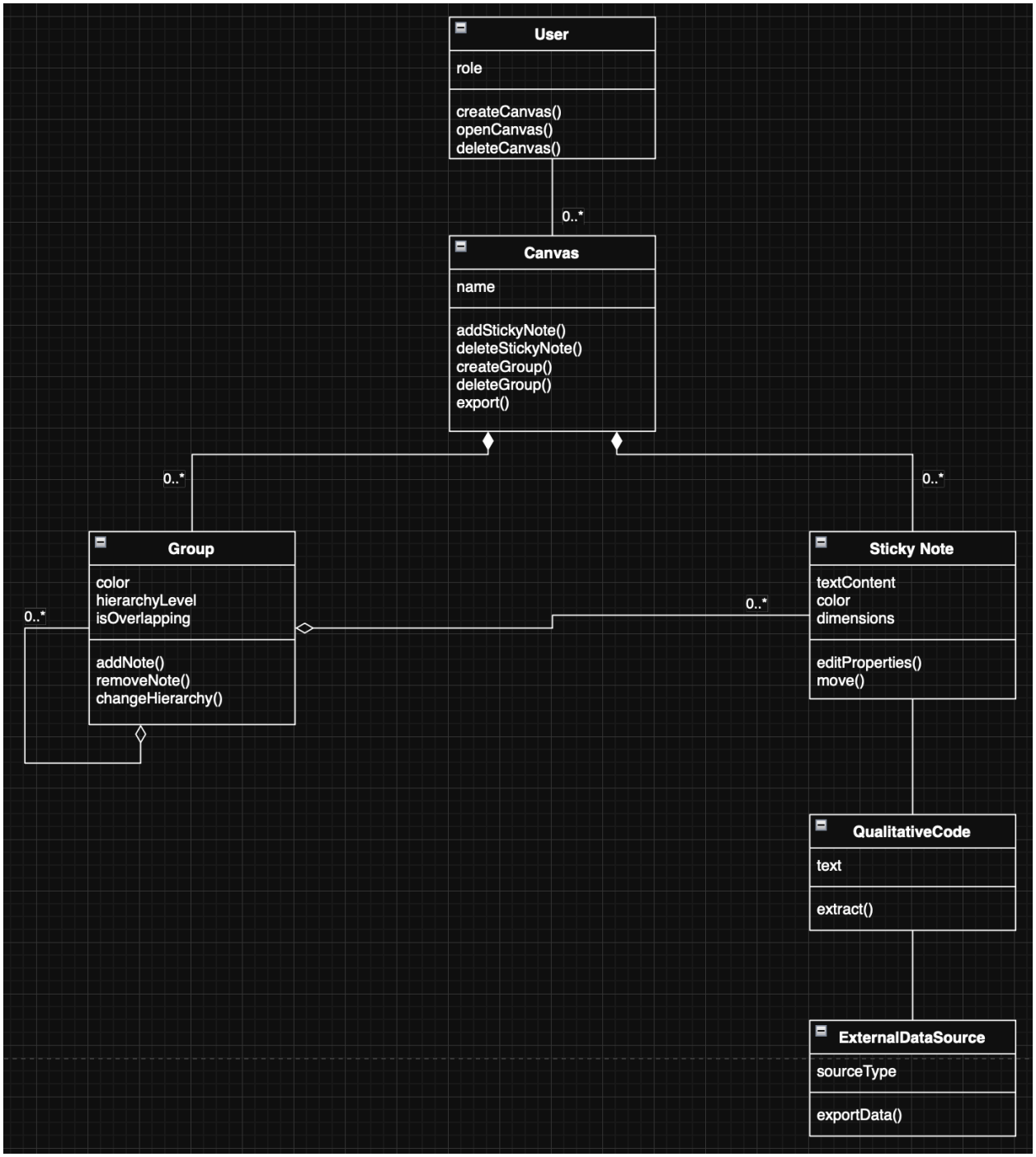
Prioritization of Backlog Items

Minimal Viable Product	1. FR1 2. Qualitative Researcher Coding, User Story 4 3. Qualitative Researcher Coding, User Story 5 4. Qualitative Researcher Coding, User Story 3 5. Qualitative Researcher Coding, User Story 2
Secondary Items	1. Qualitative Researcher Coding, User Story 1 2. Qualitative Researcher Coding, User Story 6 3. Peer Reviewer, User Story 1 4. Peer Reviewer, User Story 2 5. Peer Reviewer, User Story 3

	6. Peer Reviewer, User Story 5
Low Priority	1. Peer Reviewer, User Story 4 2. Qualitative Researcher Coding, User Story 7 3. FR15 4. FR13

Section 4: Analysis Object Model

UML Class Diagram



Section 4: Reflection

Reflections

Alejandro Ciuba: Revised and finalized *Requirements*, *User Epics*, and *Backlog*.
Milan Knezevic: Revised and finalized *Functional Requirements*. Contributed to the *Analysis Object Model* Diagram and *MVP* definition.
Vaibhav Singh: Revised and finalized *Analysis Object Model*.
Daniel Yates: Revised *Backlog* and *MVP* definition.